

SIMATS ENGINEERING
CO2_ASSESSMENT TOOL 1: Analytical Problem Solving

COURSE NAME: Cloud Computing and
Big Data Analytics

COURSE CODE: CSA1522

Name	Atmakuri Sai Mrudula
Register Number	192372087

COURSE OUTCOME COVERED

CO2: *Analyze virtualization architectures to identify performance and security issues in cloud computing environments. (BL3)*

Dave's Taxonomy: Manipulation (Level 2)

Total Marks: 50

Question Count: 5

Code of Conduct

I Atmakuri Sai Mrudula (192372087) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: A. Sai Mrudula

Assessment Tasks

Analytical Problem Solving

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission-critical applications. Analyze performance, security, and manageability, then justify the better choice.
2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.
3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.
4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.
5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Analytical Problem Solving Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 – Proficient	Level 2 - Developing	Level 1 - Beginning
Problem Interpretation	20%	Interprets all technical requirements accurately	Interprets most requirements correctly	Partial interpretation	Misinterprets the problem
Analytical Reasoning	25%	Strong reasoning with clear cause-effect analysis	Good reasoning with minor gaps	Basic analysis only	Weak or no analysis
Method Selection	20%	Chooses highly appropriate virtualization and security concepts	Chooses mostly suitable concepts	Partially suitable concepts	Inappropriate approach
Solution Quality	20%	Provides feasible and technically sound solution	Reasonable solution with minor issues	Basic solution with limitations	Unclear or invalid solution
Presentation of Analysis	15%	Well-structured and technically precise	Generally clear	Somewhat unclear	Difficult to follow

Total Marks Awarded:

1. Compare Type-1 and Type-2 hypervisors for a cloud datacenter hosting mission - critical applications.

Analyze performance, security, and manageability, then justify the better choice.

Introduction: -

A hypervisor is virtualization software that creates and manages virtual machines. It allows multiple operating systems to run on a single physical server by sharing hardware resources. Hypervisors are classified into Type-1 is Bare-Metal and Type-2 is Hosted. The choice of hypervisor is important for mission-critical applications such as banking, healthcare, and government systems, where high performance, security, and reliability are essential.

Type-1 Hypervisor: -

A Type-1 hypervisor is installed directly on the physical hardware without requiring a host operating system. It has direct access to hardware resources, making it faster and more efficient. It is commonly used in enterprise cloud datacenters. Examples include VMware ESXi, Microsoft Hyper-V, Xen, and KVM.

Type-2 Hypervisor: -

A Type-2 hypervisor is installed on top of an existing operating system. The host operating system manages the hardware, while the hypervisor manages the virtual machines. It is mainly used for software testing, development, and educational purposes. Examples include Oracle VirtualBox and VMware Workstation.

Performance Analysis: -

Type-1 hypervisors provide better performance because they access the hardware directly, reducing overhead and improving CPU, memory, and storage utilization. They support large numbers of virtual machines with minimal latency.

Type-2 hypervisors have lower performance because every hardware request passes through the host operating system. This additional layer increases processing time and reduces efficiency.

Security Analysis: -

Type-1 hypervisors offer stronger security because they have a smaller attack surface and better isolation between virtual machines. They support advanced security features such as Secure Boot, Trusted Platform Module, and VM encryption.

Type-2 hypervisors depend on the security of the host operating system. If the host operating system is compromised, all virtual machines running on it may also be affected.

Manageability Analysis: -

Type-1 hypervisors include advanced management features such as centralized administration, live migration, high availability, automatic load balancing, backup, and disaster recovery. These features make them suitable for large cloud environments.

Type-2 hypervisors provide only basic management features and are intended for small-scale environments rather than enterprise cloud datacenters.

Justification: -

For mission-critical cloud applications, Type-1 hypervisors are the better choice because they provide high performance, stronger security, efficient resource management, and enterprise-level scalability. They ensure continuous availability and better reliability for business-critical services.

Conclusion: -

Although both hypervisors support virtualization, Type-1 hypervisors are more suitable for cloud datacenters hosting mission-critical applications due to their superior performance, security, and management capabilities.

2. A cloud provider must isolate workloads belonging to finance, healthcare, and student portals on the same hardware. Analyze how virtualization architecture supports isolation and identify two major attack surfaces.

Introduction: -

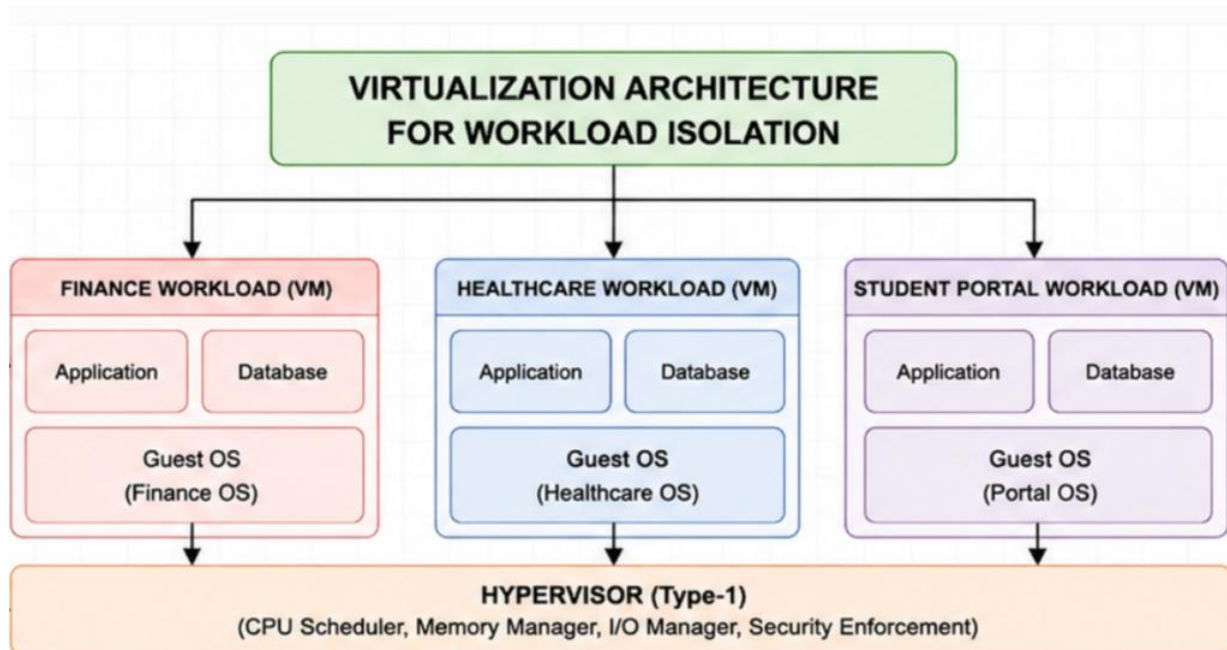
Cloud providers use virtualization to host multiple customers on the same physical server. This is called multi-tenancy. Even though the hardware is shared, each customer's applications and data must remain isolated to ensure security and privacy.

Workload Isolation Using Virtualization: -

The hypervisor creates separate virtual machines for each workload. Every virtual machine has its own operating system, CPU allocation, memory, storage, and network configuration. This prevents one application from interfering with another.

CPU isolation ensures that each virtual machine receives a fair share of processor resources. Memory isolation prevents one virtual machine from accessing the memory of another. Storage isolation provides separate virtual disks and encrypted storage for each workload. Network isolation uses VLANs, Software Defined Networking (SDN), and virtual firewalls to separate network traffic. Access control mechanisms such as Identity and Access Management (IAM), Role-Based Access Control (RBAC), and Multi-Factor Authentication (MFA) restrict access to authorized users.

Virtualization uses a hypervisor to create multiple virtual machines (VMs) on a single physical server. Each VM has its own operating system, applications, virtual CPU, memory, storage, and network configuration. This ensures that one workload cannot directly access or interfere with another. CPU isolation ensures that each VM receives a dedicated share of processor resources. The hypervisor schedules CPU time fairly among all virtual machines so that no single workload consumes excessive processing power.



Major Attack Surfaces: -

The first major attack surface is VM Escape. In this attack, a hacker exploits a vulnerability in the hypervisor and gains access to the host operating system. Once the host is compromised, other virtual machines on the same server may also be attacked.

The second major attack surface is the Side-Channel Attack. This attack exploits shared hardware resources, such as CPU cache or memory timing, to collect sensitive information like passwords, encryption keys, or confidential data.

Security Measures: -

Cloud providers reduce these risks by regularly updating hypervisors, enabling Secure Boot, encrypting data, implementing network segmentation, and continuously monitoring the infrastructure for suspicious activity.

Conclusion: -

Virtualization architecture provides secure isolation of CPU, memory, storage, network, and user access. Proper security measures help protect against VM escape and side-channel attacks, ensuring secure operation of multi-tenant cloud environments.

3. Study the following VM host utilization snapshot and recommend corrective actions: CPU 92%, memory 88%, storage I/O latency 35 ms, average VM boot time 170 s. Explain the likely bottlenecks.

Introduction: -

The given utilization snapshot shows the current performance of a virtual machine host. High utilization values indicate that the host is under heavy load and may experience reduced performance. The recorded values are CPU utilization of 92%, memory utilization of 88%, storage I/O latency of 35 milliseconds, and an average VM boot time of 170 seconds.

CPU Analysis: -

A CPU utilization of 92% indicates that the processor is almost fully occupied. This may result from too many virtual machines running simultaneously or CPU overcommitment. High CPU usage increases response time and reduces application performance.

Memory Analysis: -

Memory utilization of 88% indicates limited available RAM. When physical memory becomes insufficient, the system may start swapping data to disk, causing slower application execution and reduced VM performance.

Storage Analysis: -

A storage I/O latency of 35 milliseconds is significantly higher than the recommended value. This suggests that the storage system is overloaded or using slow disks. High storage latency delays data access and affects application performance.

VM Boot Time Analysis: -

A VM boot time of 170 seconds is much longer than expected. This is mainly caused by high CPU usage, insufficient memory, and slow storage performance. As a result, virtual machines require more time to initialize and become operational.

Recommended Corrective Actions: -

The cloud administrator should increase CPU and memory resources by upgrading the host or distributing workloads across additional servers. Live migration can move virtual machines to less-loaded hosts without interrupting services. Storage performance can be improved by replacing traditional hard disks with SSD or NVMe storage devices. Removing unused virtual machines and optimizing application resource usage will further improve performance. Continuous monitoring tools should also be used to detect resource bottlenecks early and trigger automatic scaling when necessary.

Expected Benefits: -

Implementing these corrective actions will reduce CPU and memory utilization, lower storage latency, improve application response time, and decrease VM boot time. Overall system reliability and user experience will also improve.

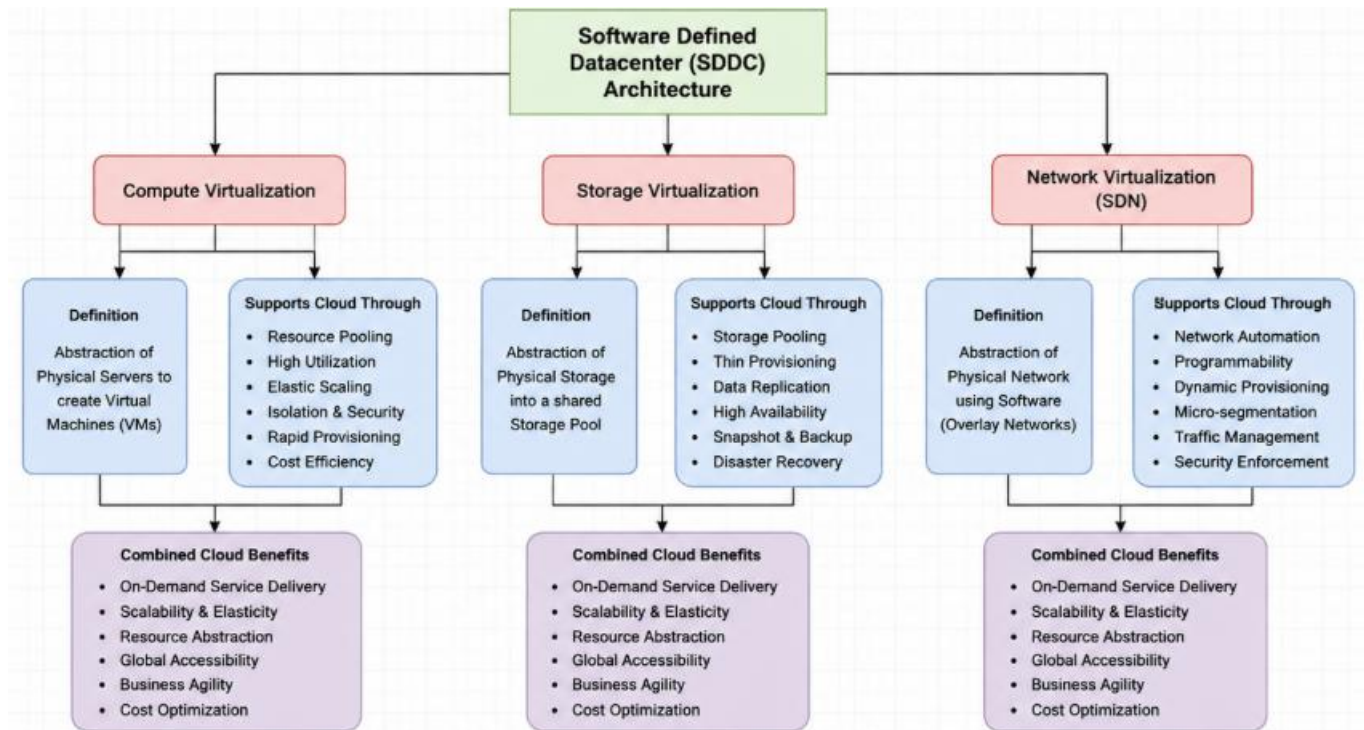
Conclusion: -

The major bottlenecks are CPU saturation, memory pressure, and storage latency. Upgrading hardware resources and distributing workloads more effectively will significantly improve system performance and reduce VM boot time.

4. Analyze how Software Defined Datacenter architecture improves agility compared with a traditional datacenter. Support your answer with compute, storage, and network virtualization considerations.

Introduction: -

A Software Defined Datacenter (SDDC) is a modern cloud infrastructure in which compute, storage, and network resources are virtualized and managed through software. Unlike traditional datacenters that rely heavily on physical hardware and manual configuration, an SDDC automates resource management and provides greater flexibility, scalability, and efficiency.



Compute Virtualization: -

In a traditional datacenter, applications are deployed on dedicated physical servers. Adding or upgrading resources often requires manual installation and configuration. In contrast, SDDC uses compute virtualization to create multiple virtual machines on a single physical server.

Storage Virtualization: -

Storage virtualization combines multiple physical storage devices into a single logical storage pool. Administrators can allocate storage dynamically without depending on specific hardware. Features such as snapshots, replication, thin provisioning, backup, and disaster recovery improve storage efficiency and simplify data management. Organizations can expand storage capacity without interrupting existing services.

Network Virtualization: -

Network virtualization is implemented using Software Defined Networking (SDN). Instead of manually configuring physical switches and routers, administrators manage networks through software. Virtual switches, routers, firewalls, and load balancers can be created quickly, making network deployment faster and more flexible. Network virtualization also supports micro-segmentation, which improves security by isolating workloads.

Agility Improvements: -

An SDDC significantly improves agility by automating resource provisioning and management. New virtual machines and services can be deployed within minutes instead of days. Resources can be allocated dynamically based on workload requirements, reducing idle hardware and improving utilization. Centralized management allows administrators to control compute, storage, and network resources from a single interface. Automation reduces manual errors, lowers operational costs, and accelerates application deployment.

Conclusion: -

Compared with traditional datacenters, Software Defined Datacenter architecture offers greater agility through compute, storage, and network virtualization. As a result, SDDC has become the preferred architecture for modern cloud environments.

5. A multi-cloud federation project fails due to identity mismatches between providers. Analyze the issue and propose a secure identity and privacy design using federation concepts.

Introduction: -

Multi-cloud federation enables users to access services offered by multiple cloud providers using a common identity. However, differences in authentication systems and identity formats may cause identity mismatches, resulting in login failures and inconsistent access control.

Organizations can also implement the Principle of Least Privilege (PoLP), where users receive the minimum permissions necessary to perform their tasks. This reduces the risk of accidental or malicious misuse of cloud resources.

Multi-Factor Authentication (MFA) should be enabled by requiring users to verify their identity using two or more authentication methods, such as a password, a one-time password (OTP), fingerprint, or mobile authentication application.

Identity Mismatch Issues: -

Each cloud provider may use different authentication methods, user account formats, and authorization policies. These differences can create duplicate user identities, inconsistent permissions, and difficulties in managing users across multiple cloud platforms.

Secure Identity Design: -

A secure solution is to implement Federated Identity Management (FIM) with a trusted Identity Provider (IdP). Users authenticate once through the Identity Provider, which securely shares identity information with all participating cloud providers.

Privacy Protection: -

User information should be encrypted during storage and transmission. Only the minimum required identity attributes should be shared between providers. Regular security audits, logging, and compliance monitoring further protect user privacy.

Benefits of Identity Federation: -

Identity federation provides several advantages. It simplifies user management by maintaining a single digital identity across multiple cloud providers. Single Sign-On improves the user experience by reducing the need for multiple logins. Standard authentication protocols ensure compatibility between different cloud platforms. Strong authentication, encryption, and role-based access control improve overall security. Centralized identity management also reduces administrative effort and supports regulatory compliance.

Conclusion: -

Identity mismatches can prevent successful multi-cloud integration. By implementing Federated Identity Management, Single Sign-On, standard authentication protocols, RBAC, MFA, and encryption, organizations can provide secure, reliable, and privacy-preserving access across multiple cloud providers.